MERN STACK

TRAINING TR-103 REPORT DAY 16

13 July 2026

Overview:

The sixteenth day of the MERN Stack training introduced the fundamentals of ReactJS. I learned about React, its key features, and the process of setting up a React application using Create React App. The session also covered the React project structure, reusable components, and the use of import and export statements for organizing code. Practical exercises helped us understand how React applications are created and managed

Learning Objectives:

- Understand the basics of ReactJS.
- Learn the features of React.
- Set up a React application using Create React App.
- Understand the React project structure.
- Learn about React components.
- Understand the use of import and export statements.

REACT

React is a JavaScript library used to build fast, interactive, and reusable user interfaces. It follows a component-based architecture, making web applications easier to develop and maintain.

Features:

- Component-based architecture.
- Virtual DOM for better performance.
- Reusable UI components.
- One-way data flow.
- Uses JSX syntax.

```
function App() {

      return (
            <p>Jeeya this side! </p>
       );

}

export default App;
```

React App Setup

A React application can be created using the Create React App tool, which automatically sets up the required files and dependencies.

Steps:

- Install Node.js.
- Verify installation using node -v and npm -v.

- Create a React project using:

```
node –v
npm –v
npx -v
```

```
npx create-react-app my-app
```

Benefits

- Quick project setup.
- Automatically installs required packages.
- Provides a ready-to-use React environment.

React Project Structure

A React project contains folders and files that organize the application efficiently.

Important Folders:

- src – Contains source code, components, CSS, and assets.
- public – Stores static files.
- node_modules – Contains installed packages.

Benefits:

- Improves code organization.
- Makes applications easier to maintain.
- Encourages code reusability.

React Components

A React Component is a reusable and independent part of the user interface. Components help divide a large application into smaller and manageable sections.

Features:

- Reusable and modular.
- Improves code readability.
- Simplifies application development.
- Functional components are commonly used.

Import

The import statement is used to bring components, functions, or modules from one file into another.

Features:

- Reuses code across files.
- Improves project organization.
- Usually written at the top of a file.

Export

The export statement makes components, functions, or variables available for use in other files.

Types of Export:

- Default Export – One default export per file.
- Named Export – Multiple named exports in a file.

Benefits:

- Promotes code reusability.
- Helps organize React applications.

TASK :

```jsx
export const Navbar =() =>{
    return(
        <>
            <nav className="navbar navbar-expand-lg bg-body-tertiary">
  <div className="container-fluid">
    <a className="navbar-brand" href="#">Navbar</a>
    <button className="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarSupportedContent" aria-controls="navbarSupportedContent" aria-expanded="false" aria-label="Toggle navigation">
      <span className="navbar-toggler-icon"></span>
    </button>
    <div className="collapse navbar-collapse" id="navbarSupportedContent">
      <ul className="navbar-nav me-auto mb-2 mb-lg-0">
        <li className="nav-item">
          <a className="nav-link active" aria-current="page" href="#">Home</a>
        </li>
        <li className="nav-item">
          <a className="nav-link" href="#">Link</a>
        </li>
        <li className="nav-item dropdown">
          <a className="nav-link dropdown-toggle" href="#" role="button" data-bs-toggle="dropdown" aria-expanded="false">
            Dropdown
          </a>
          <ul className="dropdown-menu">
            <li><a className="dropdown-item" href="#">Action</a></li>
            <li><a className="dropdown-item" href="#">Another action</a></li>
            <li><hr className="dropdown-divider"/></li>
            <li><a className="dropdown-item" href="#">Something else here</a></li>
          </ul>
        </li>
        <li className="nav-item">
          <a className="nav-link disabled" aria-disabled="true">Disabled</a>
        </li>
      </ul>
      <form className="d-flex" role="search">
        <input className="form-control me-2" type="search" placeholder="Search" aria-label="Search"/>
        <button className="btn btn-outline-success" type="submit">Search</button>
      </form>
    </div>
  </div>
</nav>


        </>
    )
}
```

Output :



Conclusion:

By the end of Day 16, I gained a strong understanding of the fundamentals of ReactJS. I learned how to set up a React project, organize files using the standard project structure, create reusable components, and use import and export statements to build well-structured and maintainable React applications.